

Name:

Qousain Haider

Roll-No:

2024-CET-13

Department:

Software Engineering

Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL DDL Commands (Database & Table Creation)

Submitted To:

Engr. Asma Javid

Lab Task 4

Tasks (Practice Work)

1.Create Database:

Create a database named:

college_db

Server: 127.0.0.1:3307 » Database: college_db

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Table name: employees Add 1 column(s) Go

Name	Type	Length/Values	Default	Collation	Attributes	Null	Index
	INT		None			☐	---
	INT		None			☐	---
	INT		None			☐	---
	INT		None			☐	---
	INT		None			☐	---

Table comments: Collation: Storage Engine: InnoDB

2.Create Table:

Create a table named employees with following fields:

Column Name	Data Type	Constraint
emp_id	INT	PRIMARY KEY, AUTO_INCREMENT
emp_name	VARCHAR(50)	NOT NULL
salary	DECIMAL(10,2)	DEFAULT 25000
department	VARCHAR(30)	
join_date	DATE	DEFAULT CURRENT_DATE

The screenshot shows the phpMyAdmin web interface. The top navigation bar includes links for Structure, SQL, Search, Query, Export, Import, Operations, Privileges, Routines, Events, and More. The left sidebar shows a tree view of databases, with 'college_db' selected. The main area displays the SQL query to create the 'employees' table:

```
1 CREATE TABLE employees
2 (
3   emp_id INT AUTO_INCREMENT PRIMARY KEY,
4   emp_name VARCHAR(50) NOT NULL,
5   salary DECIMAL(10,2) DEFAULT 25000,
6   department VARCHAR(30),
7   join_date DATE DEFAULT CURRENT_DATE
8 );
```

Below the query editor are buttons for 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters'. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is at the bottom right.

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

SELECT * FROM `employees`

☐ Profiling

[\[Edit inline \]](#)

[\[Edit \]](#)

[\[Explain SQL \]](#)

[\[Create PHP code \]](#)

[\[Refresh \]](#)

emp_id

emp_name

salary

department

join_date

Query results operations

Create view

Console

3.Insert Data:

Insert at least 5 records into employees.

The screenshot shows the phpMyAdmin web interface. The left sidebar displays a database structure tree with 'college_db' selected and 'employees' highlighted. The main panel is titled 'Run SQL query/queries on table college_db.employees:'. It contains an SQL editor with the following query:

```
1 INSERT INTO employees (emp_name, salary, department)
2 VALUES
3 ("Zara Khan",50000,"Software Engineering"),
4 ("Zoya Nayak",50000,"Software Engineering"),
5 ("Bilal Khan",55000,"Software Engineering"),
6 ("Saifullah Khan",530000,"Electrical Engineering"),
7 ("Toseef Khan",100000,"Medical Engineering");
8
```

Below the editor are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters'. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is on the right.

On the right side of the SQL editor, a table structure preview is visible with the following columns:

emp_id
emp_name
salary
department
join_date

At the bottom left, there is a 'Console' tab.

phpMyAdmin

Recent

Favorites

New

collegedb

college_db

New

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

SELECT * FROM `employees`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

emp_id

emp_name

salary

department

join_date

1

Zara Khan

50000.00

Software Engineering

2026-04-02

2

Zoya Nayak

50000.00

Software Engineering

2026-04-02

3

Bilal Khan

55000.00

Software Engineering

2026-04-02

4

Saifullah Khan

530000.00

Electrical Engineering

2026-04-02

5

Toseef Khan

100000.00

Medical Engineering

2026-04-02

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Console

4. Modify Table:

1. Add column email

2. Add column phone

3. Set email as **UNIQUE**

The screenshot shows the phpMyAdmin interface for the 'college_db' database, specifically for the 'employees' table. The left sidebar shows the database structure with 'college_db' selected and 'employees' highlighted. The main area displays the 'SQL' tab with the following queries:

```
1 ALTER TABLE employees ADD email VARCHAR(15) UNIQUE;
2 ALTER TABLE employees ADD phone VARCHAR(15);
3
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present. At the bottom, there is a 'Delimiter' dropdown set to ';', and several checkboxes: 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). A 'Go' button is at the bottom right.

The right sidebar shows the table structure for 'employees' with columns: emp_id, emp_name, salary, department, and join_date.

phpMyAdmin

Recent Favorites

New

- collegedb
- college_db
 - New
 - employees
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 4 (5 total, Query took 0.0007 seconds.)

```
SELECT * FROM `employees`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	emp_id	emp_name	salary	department	join_date	email	phone
☐ Edit ☐ Copy ☐ Delete	1	Zara Khan	50000.00	Software Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	2	Zoya Nayak	50000.00	Software Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	3	Bilal Khan	55000.00	Software Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	4	Saifullah Khan	530000.00	Electrical Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	5	Toseef Khan	100000.00	Medical Engineering	2026-04-02	NULL	NULL

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

phpMyAdmin

Recent Favorites

New

- collegedb
- college_db
 - New
 - courses
 - employees
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 5 (6 total, Query took 0.0010 seconds.)

```
SELECT * FROM `employees`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	emp_id	emp_name	salary	department	join_date	email	phone
☐ Edit ☐ Copy ☐ Delete	1	Zara Khan	50000.00	Software Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	2	Zoya Nayak	50000.00	Software Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	3	Bilal Khan	55000.00	Software Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	4	Saifullah Khan	530000.00	Electrical Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	5	Toseef Khan	100000.00	Medical Engineering	2026-04-02	NULL	NULL
☐ Edit ☐ Copy ☐ Delete	6	Muhamamd Bilal	25000.00	Computer Engineering	2026-04-02	muhamamdilal@g	305982154

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

5. View Structure:

Use:

DESCRIBE employees;

The screenshot shows the phpMyAdmin web interface. The left sidebar displays a database tree with 'college_db' selected, and the 'employees' table chosen. The main panel shows the 'Structure' tab for the 'employees' table. A message indicates the SQL query 'DESCRIBE employees;' was executed successfully. Below this, a table lists the columns and their properties.

Field	Type	Null	Key	Default	Extra
emp_id	int(11)	NO	PRI	NULL	auto_increment
emp_name	varchar(50)	NO		NULL	
salary	decimal(10,2)	YES		25000.00	
department	varchar(30)	YES		NULL	
join_date	date	YES		curdate()	
email	varchar(15)	YES	UNI	NULL	
phone	varchar(15)	YES		NULL	

Below the table, there are buttons for 'Print', 'Copy to clipboard', and 'Create view'. At the bottom, there is a 'Console' tab.

6. Extra Challenge (optional):

Create table courses:

course_id, course_name, duration, fee

With proper data types and constraints.

Server: 127.0.0.1:3307 » Database: college_db » Table: employees

Run SQL query/queries on table college_db.employees:

```
1 CREATE TABLE courses
2 (
3   course_id INT AUTO_INCREMENT PRIMARY KEY,
4   course_name VARCHAR(30) NOT NULL,
5   duration VARCHAR(20) NOT NULL,
6   fee INT DEFAULT 25000
7 );
```

emp_id
emp_name
salary
department
join_date
email
phone

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin



Recent Favorites

New

collegedb

college_db

New

courses

employees

information_schema

mysql

performance_schema

phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0011 seconds.)

SELECT * FROM `courses`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

course_id course_name duration fee

Query results operations

Create view

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - employees
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table college_db.courses:

```
1 INSERT INTO courses(course_name,duration,fee) VALUES
2 ("WEB Development","6 Months",50000);
```

course_id
course_name
duration
fee

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query <<

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Console

phpMyAdmin

Recent Favorites

- New
- collegedb
- college_db
 - New
 - courses
 - employees
- information_schema
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: college_db » Table: courses

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 0 (1 total, Query took 0.0018 seconds.)

SELECT * FROM `courses`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	course_id	course_name	duration	fee
☐ Edit ☐ Copy ☐ Delete	1	WEB Development	6 Months	50000

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Console